A Synchronization Critic Improves Video-to-Audio Timing at Sampling Time

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Abstract

Video-to-audio (V2A) generators usually produce the right kind of sound for a scene, but often at the wrong moment. We study how a synchronization critic can improve the timing of a codebook V2A generator (SpecVQGAN). The critic is a small audio-visual model, trained contrastively on real clips to find the offset between audio and video and then frozen. Its score, the Perceptual Alignment Loss (PAL), is the share of the critic's offset distribution that falls inside a window around zero offset, a simplified stand-in for the human audio-visual temporal binding window. We compare two ways to use PAL. As a differentiable training loss, passed to the transformer's logits through straight-through codebook decoding, PAL makes timing worse on FlashDot, a synthetic benchmark of flashing dots and tones with frame-exact labels. Two diagnostics suggest why: teacher-forced predictions inherit their alignment from the ground-truth prefix, so they look aligned without using the video, and optimizing free logits against the frozen critic reaches the score of real audio while the decoded audio stays poorly timed. As a reward on the generator's own free-running samples, the same critic helps: keeping the best of several samples improves timing on the FlashDot test split in every seed and on the open-domain VAS dataset. PAL-filtered self-training, which fine-tunes the generator on its own best-scored samples, does not keep this gain: it does no better than fine-tuning on randomly chosen samples, and it does not improve timing over the model it starts from. We also find that envelope lag recovers known delays exactly, while DTW distance tolerates time shifts by construction and is unsuited as a timing metric. In our experiments a synchronization critic improved the timing of a discrete-token V2A model only when it chose among samples at generation time. Code and FlashDot: `https://github.com/sameraslan/PerceptAlign`.

1 Introduction

Video-to-audio (V2A) generation produces a soundtrack that fits the frames of a silent clip, for uses such as Foley for film and sound for generated video. Codebook-transformer models (Iashin and Rahtu, 2021) and their successors usually get the kind of sound right. Timing is harder: a hammer clip can receive convincing hammer sounds whose impacts land between the visible strikes. Part of the reason is the training objective. Cross-entropy over discrete spectrogram tokens has no notion of temporal distance, so an onset placed slightly early costs about as much as one placed a second early.

Human perception handles timing with some tolerance. Light and sound from one event arrive at slightly different times, and observers still bind them into one event when the asynchrony falls inside the temporal binding window (Wallace and Stevenson, 2014). The window is wider when audio lags than when it leads (Dixon and Spitz, 1980; van Wassenhove et al., 2007; Vroomen and Keetels, 2010), varies between people (Stevenson et al., 2012), and shifts after exposure to a constant lag (Fujisaki et al., 2004). We take one design principle from this literature: a timing score should accept small offsets around zero and penalize large ones, rather than treating every non-zero offset as equally wrong. Calling the score "perceptual" is an analogy; we do not model the perceptual system.

We build such a score from a frozen, contrastively pretrained synchronization critic. For each short segment of a clip, the critic compares audio and video embeddings at every offset in a small range and turns the comparisons into a distribution over offsets; the Perceptual Alignment Loss (PAL) is the negative log of the mass inside a window around zero. The form follows the self-synchronization loss of Niu and Mak (2023), with a soft window over near-zero offsets in place of their one-hot target, an optional asymmetric window, and computation per segment.

A frozen critic can enter a generator in two ways. It can be a differentiable training loss, as the sync expert of Wav2Lip (Prajwal et al., 2020) is for lip generation; for a discrete-token generator this needs a path from the logits to a spectrogram, which we provide with straight-through decoding through the frozen codebook decoder. Or it can be a

reward on whole samples from the generator, as in re-ranking (Du et al., 2023) and preference optimization (Chan et al., 2026; Cheng et al., 2026). We study both on SpecVQGAN and find a difference. As a loss, PAL does not improve timing, and two simple probes point to possible reasons. Teacher-forced predictions are conditioned on the aligned ground-truth prefix, so they look aligned to the critic whether or not the model uses the video (exposure bias, Bengio et al., 2015; Ranzato et al., 2016). And when the gradient path is opened up so that only the video can determine the timing, optimizing logits against the frozen critic finds spectrograms that it scores as well aligned but that do not follow the video. As a reward on free-running samples, the critic avoids both problems: candidates are realistic outputs of the generator, so the critic only has to rank them. Choosing the best of several samples this way improves timing on FlashDot and lowers the envelope lag on VAS, where the generator's timing stays near chance, while fine-tuning the generator on the chosen samples does no better than fine-tuning on randomly chosen ones. Because a timing metric that compares spectrograms can be dominated by spectral content, we also test candidate metrics against known delays on a benchmark where the correct timing is known exactly.

Contributions.

- **PAL**, a windowed, per-segment synchronization score from a frozen critic, used as a loss and as a reward (Section 3).
- **An analysis of PAL's failure as a loss** for a codebook generator, with two reusable diagnostics that are consistent with it: a wrong-video cross-entropy probe for exposure bias and a free-logit attack that tests whether a critic can be exploited (Sections 3.6 and 5.2).
- **PAL-filtered self-training**, which fine-tunes the generator on its own lowest-PAL samples, with a random-selection control on both datasets and a ground-truth-only control on VAS, alongside best-of-N re-ranking (Sections 3.5 and 5.3).
- **FlashDot**, a synthetic benchmark of flashing dots and synchronous tones with frame-exact timing labels, and **a sanity check of timing metrics** against known delays (Sections 4.1 and 5.1).

2 Related work

Video-to-audio generation. Early systems include example-based synthesis on Greatest Hits (Owens et al., 2016), waveform generation on VEGAS (Zhou et al., 2018) and RegNet on VAS (Chen et al., 2020). SpecVQGAN (Iashin and Rahtu, 2021), our base model, adapts VQGAN (Esser et al., 2021) to spectrograms and predicts codes from visual features with a transformer. Other systems improve alignment through architecture or visual features (Mei et al., 2024; Wang et al., 2024c; Viertola et al., 2025; Cheng et al., 2025; Mo and Song, 2025), or condition on a timing signal predicted from video such as onsets, energy or a coarse layout (Comunità et al., 2024; Zhang et al., 2026; Ton et al., 2025; Jeong et al., 2025; Wang et al., 2024b). Semantic conditioning without a timing mechanism is also common (Sheffer and Adi, 2023; Wang et al., 2024a).

Synchronization signals for V2A. Several systems use a synchronization model on whole outputs. CondFoley (Du et al., 2023) re-ranks samples from a code model with a synchronization model at test time. Diff-Foley (Luo et al., 2023) guides diffusion sampling with an alignment classifier, and Seeing and Hearing (Xing et al., 2024) steers pretrained generators with a frozen multimodal aligner. V2A-DPO (Chan et al., 2026) and SyncDPO (Cheng et al., 2026) post-train with preference pairs that reward synchronization. Our self-training is in the same family of sample-level rewards, with a binding-window critic as the reward and supervised fine-tuning on selected samples in place of a preference objective.

Synchronization models and losses. SyncNet (Chung and Zisserman, 2017) learned a two-stream embedding for lip sync. Owens and Efros (2018) learned audio-visual features by detecting shifted audio, the supervision our critic uses, and Chung et al. (2019) trained with a softmax over many candidate offsets, the form of our critic objective. Later models cover lips and voices (Kadandale et al., 2022; Chen et al., 2024), localization (Afouras et al., 2020; Arandjelović and Zisserman, 2018) and open-domain video (Iashin et al., 2022, 2024). Inside generators, Wav2Lip (Prajwal et al., 2020) trains a face generator against a frozen sync expert, VCA-GAN (Kim et al., 2021) adds a contrastive

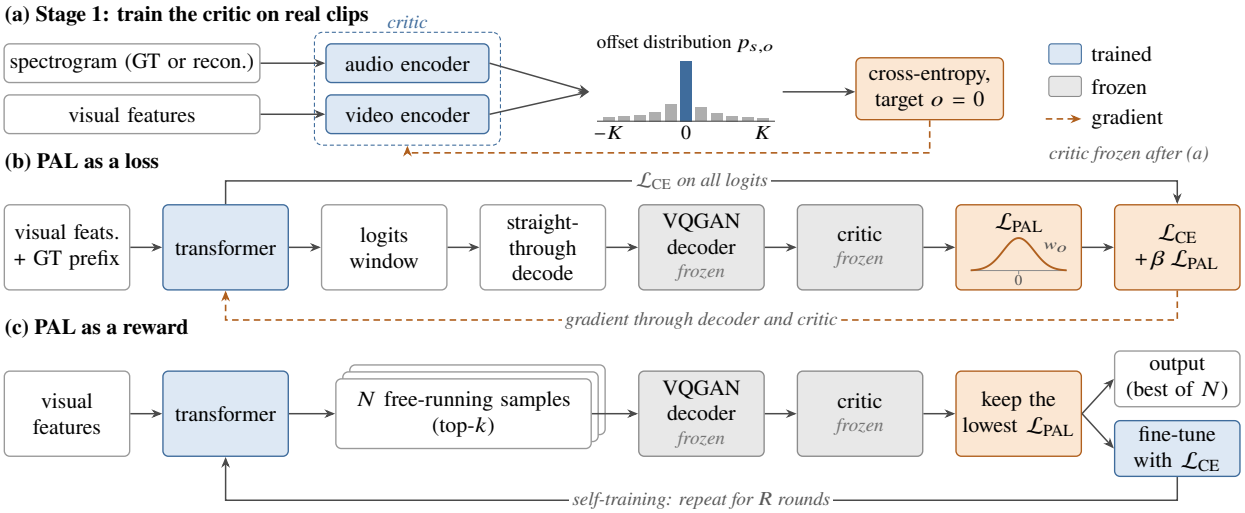


Figure 1: Two uses of one frozen critic. (a) Stage 1: the critic embeds a real spectrogram (ground truth or its VQGAN reconstruction) and the clip's visual features and is trained so that zero offset wins the softmax over offsets. (b) PAL as a loss (Section 3.4): a window of the transformer's logits is decoded with a straight-through estimator, scored by the critic, and PAL is added to the cross-entropy loss. (c) PAL as a reward (Section 3.5): free-running samples are decoded and scored, and the lowest-PAL sample is kept for output or for fine-tuning.

synchronization term to lip-to-speech, Niu and Mak (2023) use the negative log probability of zero offset from a cross-correlation over offsets, and LatentSync (Li et al., 2024) carries SyncNet supervision into latent diffusion. These methods produce continuous outputs for lips and speech; we ask whether the same kind of loss transfers to a discrete-token sound-effect generator, and find that it does not.

Exposure bias, self-training and reward over-optimization. Teacher forcing conditions each prediction on the ground-truth prefix, so training never sees the model's own errors; scheduled sampling feeds model predictions back during training (Bengio et al., 2015), and sequence-level training scores whole generated sequences (Ranzato et al., 2016). Training on a model's own filtered outputs appears in expert iteration (Anthony et al., 2017) and in ReST (Gulcehre et al., 2023), which alternate sampling from the current policy, filtering by a reward and fine-tuning on the kept samples; our self-training follows that loop with a synchronization critic as the filter. A learned reward that is optimized hard drifts away from the quantity it stands for (Gao et al., 2023); our free-logit attack measures the extreme case, with no generator between the optimizer and the critic.

Human audio-visual timing. Observers perceive auditory and visual signals as one event over a range of asynchronies, the temporal binding window (Wallace and Stevenson, 2014). Audio lead is detected at smaller asynchronies than audio lag (Dixon and Spitz, 1980), and the window for audio-visual speech is wider on the audio-lag side (van Wassenhove et al., 2007; Vroomen and Keetels, 2010). Its width differs across individuals (Stevenson et al., 2012), and perceived simultaneity recalibrates after exposure to a constant lag (Fujisaki et al., 2004; Takeshima, 2021). We use these findings only to motivate the shape of the target over offsets.

3 Method

3.1 Base generator: SpecVQGAN

We follow SpecVQGAN (Iashin and Rahtu, 2021). Audio is an 80-bin log-mel spectrogram (86.1 frames per second) of 848 frames, about 9.8 s, with values in $[0, 1]$. A VQGAN (Esser et al., 2021) encodes it into a 5×53 grid of codes from a codebook $E \in \mathbb{R}^{|V| \times 256}$, $|V| = 128$; one code column covers 16 mel frames. The transformer reads codes in column-major order (the five codes of a column, then the next column), so a sequence has $N = 265$ tokens. It is a

GPT whose input starts with the clip's 212 visual feature frames (2048-dimensional, 21.5 per second, four per code column), followed by the token embeddings. It is trained by teacher-forced cross-entropy, $\mathcal{L}_{\text{CE}} = -\frac{1}{N}\sum_j \log q_j(z_j)$ with $q_j = \text{softmax}(\ell_j)$, where z_j is the ground-truth code and $\ell_j \in \mathbb{R}^{|V|}$ the logits at position j, computed from the visual features and the ground-truth prefix $z_{<j}$. At test time codes are sampled autoregressively (top-k with $k = 100$, temperature 1) and decoded by the VQGAN decoder.

3.2 Synchronization critic

The critic has an audio encoder and a video encoder of the same structure: two blocks of a temporal one-dimensional convolution (kernel 5, 256 channels), group normalization (8 groups) and GELU, then a 1×1 convolution to 128 dimensions and L2 normalization at every frame. The audio input is the spectrogram mapped to $[-1, 1]$ and average-pooled by four along time, which brings it to the 212-frame rate of the visual features. In a band variant the audio encoder sees only $B \in \{8, 4, 1\}$ band energies, the means over equal groups of mel bins, so it can judge when energy changes but sees little of the spectral content. The design follows the local feature extractors of Niu and Mak (2023). For a clip the critic gives unit-norm embeddings $a_1, \ldots, a_T$ and $v_1, \ldots, v_T$.

3.3 Perceptual Alignment Loss

For offsets $o \in \{-K, \ldots, K\}$ (in video frames), PAL[1] compares audio frame i with video frame $i + o$, so a negative offset means the audio event comes after the visual one (audio late). The central frames $i \in [K, T - K)$ are split into S equal consecutive segments $I_1, \ldots, I_S$, so every offset compares the same number of real frames. For segment s,

$$C_{s,o} = \frac{1}{|I_s|}\sum_{i \in I_s} \langle a_i, v_{i+o} \rangle, \qquad p_{s,o} = \frac{\exp(C_{s,o}/\tau)}{\sum_{o'=-K}^{K} \exp(C_{s,o'}/\tau)}. \tag{1}$$

The target is a window over offsets with weight one at zero,

$$w_o = \exp\left(-\frac{o^2}{2\sigma_o^2}\right), \quad \sigma_o = \begin{cases} \sigma_{\text{late}} & o < 0 \text{ (audio late)}, \\ \sigma & o \geq 0, \end{cases} \qquad W_s = \sum_{o=-K}^{K} w_o \, p_{s,o} \in (0, 1]. \tag{2}$$

The symmetric window has $\sigma_{\text{late}} = \sigma$; the asymmetric option sets $\sigma_{\text{late}} > \sigma$, widening the window on the audio-late side in the direction of the human asymmetry. For $\sigma_o = 0$ we take w_o to be the indicator of $o = 0$, so $\sigma = 0$ scores exact zero offset only. PAL averages the windowed mass over segments and takes the negative logarithm:

$$\mathcal{L}_{\text{PAL}} = -\log\left(\frac{1}{S}\sum_{s=1}^{S} W_s\right). \tag{3}$$

With $\sigma = 0$ and $S = 1$, Equation (3) is $-\log p_{1,0}$, the self-synchronization loss of Niu and Mak (2023). The window gives partial credit for mass one or two frames from zero, and segmentation keeps one well-aligned stretch from hiding a badly aligned one. Stage 1 uses the per-segment reduction $-\frac{1}{S}\sum_s \log W_s$.

Stage 1: training the critic. The critic is trained on real clips with $\sigma = 0$, $K = 10$, $\tau = 0.1$ and the per-segment log reduction ($S = 8$ on VAS, $S = 4$ on FlashDot), so the negatives are the other offsets of the same clip (Owens and Efros, 2018; Chung et al., 2019) and the objective is a cross-entropy over offsets with zero offset as the target in every segment. The audio input is the VQGAN reconstruction of the ground truth (the ground truth itself with probability 0.3), since every spectrogram the critic scores later comes out of the same decoder. The critic is then frozen.

3.4 PAL as a training loss

The critic needs a spectrogram, while the transformer outputs a distribution over codes. With $q_j = \text{softmax}(\ell_j)$, the decoded code vector is

$$\hat{e}_j = \left(\text{onehot}\left(\arg\max_k q_{j,k}\right) + q_j - \text{sg}(q_j)\right)^{\top} E, \tag{4}$$

[1]The name reflects the motivation from the temporal binding window; Section 6 reports that our results do not support a perceptual benefit of the window.

Algorithm 1 PAL-filtered self-training

Require: generator G, frozen decoder D and critic, training clips $\{(x_i, z_i)\}$, rounds R, clips per round M, candidates N, steps per round U

for $r = 1, \ldots, R$ **do**
  draw M training clips $\mathcal{I}$
  for $i \in \mathcal{I}$ **do**
    sample $\tilde{z}_i^{(1)}, \ldots, \tilde{z}_i^{(N)} \sim G(\cdot \mid x_i)$ ▷ free-running, top-k
    $k^\star \leftarrow \arg\min_k \mathcal{L}_{\mathrm{PAL}}\big(D(\tilde{z}_i^{(k)}), x_i\big)$ ▷ random control: $k^\star \leftarrow 1$
    $\hat{z}_i \leftarrow \tilde{z}_i^{(k^\star)}$
  end for
  for U steps **do**
    batch: half $(x_i, \hat{z}_i)$ with $i \in \mathcal{I}$, half (x_j, z_j) from all training clips
    update G (or its LoRA adapters) by teacher-forced $\mathcal{L}_{\mathrm{CE}}$ on the batch
  end for
end for

where sg stops the gradient (Bengio et al., 2013). The forward value is the codebook entry of the most likely code, so the decoder only receives real codebook entries, and the backward pass uses the gradient of the expected code vector $q_j^\top E$. The code vectors are decoded by the frozen decoder and scored by the critic. To bound memory, PAL is computed on a random window of 12 code columns (about 2.2 s, decoded with one extra column of context on each side) for four examples per batch, with $K = 8$, $S = 4$, $\tau = 0.1$ and $\sigma = 1$. The loss is $\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \beta\, \mathcal{L}_{\mathrm{PAL}}$.

Corrupted-prefix PAL. If the logits come from the teacher-forced pass, PAL scores the most likely codes given the ground-truth prefix. In column-major order the prefix contains the codes of every earlier column, which already show whether a sound is on, so these predictions can look aligned without any use of the video. We therefore also compute PAL on a second forward pass in which each input token is replaced by a uniformly random code with probability $1 - p_{\mathrm{keep}}^{\mathrm{PAL}}$. With $p_{\mathrm{keep}}^{\mathrm{PAL}} = 0$ the prefix carries no information, and the video is the only source of timing. Cross-entropy can use the same corruption on its own pass ($p_{\mathrm{keep}} < 1$), a simple form of training on imperfect prefixes in the spirit of scheduled sampling (Bengio et al., 2015).

3.5 PAL as a reward on samples

Best-of-N re-ranking. For each clip we draw N free-running samples, score each with PAL under the training critic, and output the lowest-PAL sample. This needs no training and multiplies test-time cost by N.

PAL-filtered self-training. Self-training moves the selection into training (Algorithm 1). Each round draws N samples for a random subset of training clips, keeps the lowest-PAL sample per clip, and fine-tunes the generator with cross-entropy on batches that mix kept samples and ground truth half and half. The critic only ranks realistic outputs of the generator and is never differentiated. Two controls share every setting: *random selection* keeps an arbitrary candidate (the one drawn first) in place of the lowest-PAL one, so any difference comes from the critic's choice; *ground-truth-only* fine-tuning runs the same number of steps on ground truth alone, so any difference from it comes from training on the model's own samples. On VAS only LoRA adapters (Hu et al., 2022) on the frozen SpecVQGAN transformer are trained.

3.6 Diagnostics

Two cheap probes tell, before a long training run, whether a critic-based loss can work for a given autoregressive code model and frozen critic.

Wrong-video cross-entropy. We compute development cross-entropy twice, once with each clip's own visual features and once with the features of a different development clip, and repeat both with a fully corrupted prefix ($p_{\mathrm{keep}} = 0$). The first gap shows whether the model uses the video at all; the second, and its level against uniform ($\log |V| \approx 4.85$),

whether the video alone can carry its predictions. If it cannot, a loss that needs the model to use the video has little to work with.

Free-logit attack. We replace the generator with free logits for eight development clips, initialized at random and optimized by Adam (learning rate 0.1, 200 steps) to minimize PAL through the same straight-through decoding and window as in training. If the PAL reached matches that of the ground-truth codes while the timing metrics (Section 4.3) of the decoded optimum stay near those of random codes, the critic can be satisfied by audio that does not follow the video. We hypothesize that a generator trained against it can learn the same shortcut; we did not test this directly.

4 Experimental setup

4.1 Datasets

FlashDot. FlashDot is a synthetic benchmark in which the correct timing is known exactly (example in Figure 5). Each 10 s clip shows a red dot on a white background during 1–6 "on" intervals of 0.15–1.5 s and plays a 1 kHz tone during exactly the same intervals; video runs at 21.5 frames per second, and clips are generated from a seed. Spectrograms follow the SpecVQGAN VAS pipeline, and visual features are the 2048-dimensional pooled outputs of an ImageNet-trained ResNet-50 (He et al., 2016), one per frame. We generate 2000 clips, split 1600/200/200 into train, development and test.

VAS. VAS (Chen et al., 2020) has eight classes (baby, cough, dog, drum, fireworks, gun, hammer, sneeze). We use the spectrograms and features released with SpecVQGAN (Iashin and Rahtu, 2021): BN-Inception (Ioffe and Szegedy, 2015) RGB and optical-flow features, flow from denseflow (Wang et al., 2020), concatenated to 2048 dimensions, 212 frames per clip. The test set is SpecVQGAN's `vas_valid` list (768 clips). A class-stratified 5% of `vas_train` is held out as a development split (589 clips), and the remaining 11,184 clips are used for training.

4.2 Models

On FlashDot we train a GPT with 6 layers, 6 heads and width 384 from scratch on codes from the public SpecVQGAN VAS codebook, which is not retrained (batch 8, 2000 steps unless stated). On VAS we start from the public SpecVQGAN VAS transformer (24 layers, 16 heads, width 1024). To fit training on a laptop GPU, its weights are frozen in half precision and only LoRA adapters (Hu et al., 2022) are trained, on the key, query, value and output projections and both MLP layers of every block (rank 16, $\alpha = 16$, initialized so that training starts from the public model). All training uses AdamW (Loshchilov and Hutter, 2019). Table 5 lists all hyperparameters.

4.3 Metrics

Timing metrics compare a generated spectrogram with the ground-truth spectrogram of the same clip. The onset envelope is the spectral flux of the log-mel spectrogram (the sum over bins of positive frame-to-frame differences). *Onset F1* matches generated and ground-truth onsets one to one within ±100 ms (Raffel et al., 2014). *Onset AP* ranks generated onsets by strength and counts a hit when one falls within ±100 ms of an unmatched ground-truth onset, as in Du et al. (2023). For the envelope metrics both envelopes are smoothed and mean-centred: *envelope correlation* is their Pearson correlation at zero lag, and *lag* is the shift within ±1 s that maximizes their cross-correlation, reported as mean absolute value in ms and as the share of clips *within one video frame* ($|\text{lag}| \leq 46.5$ ms).

On FlashDot, *tone F1* is the frame-level F1 between the ground-truth on/off label and a tone detector that thresholds the mean of the three mel bins with the largest on-minus-off difference. Bins and threshold are chosen on VQGAN reconstructions of training clips. Random codes score about 0.20.

Scoring a model with the critic it was selected against invites reward hacking, so *held-out sync accuracy* uses a separate evaluation critic trained with the stage-1 recipe, a different seed and a different architecture (kernel 3, three blocks, width 384; on VAS its audio input is 8 band energies). It scores the whole clip ($S = 1$, $\sigma = 0$) over 21 offsets; we report the fraction of clips whose most likely offset is zero (tolerance 0) or within one frame of zero (tolerance 1). Chance is about $1/21$ and $3/21$. The critics are weak on VAS: development offset accuracy is 0.116 (0.288 within one frame) for the training critic and 0.124 (0.311) for the held-out critic. On FlashDot the training critic reaches 0.78

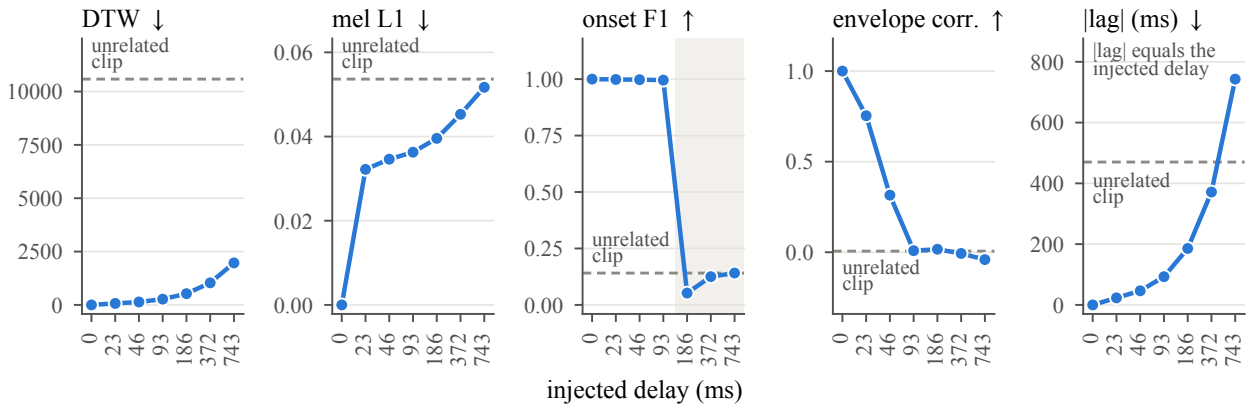


Figure 2: Metric response to injected delays on FlashDot (development, 200 clips). Dashed lines: an unrelated clip. Only |lag| grows in proportion to the delay; DTW and mel L1 rank every delay as closer to the truth than an unrelated clip.

(0.81), limited by silent segments, which have no offset to find. On VAS we also report Melception FID and mean KL divergence (MKL) as in SpecVQGAN (Iashin and Rahtu, 2021). The metric check (Section 5.1) adds DTW distance on per-clip standardized mel frames (Salvador and Chan, 2007) and mean absolute mel difference (mel L1).

4.4 Protocol

All settings are chosen on the development split, and each final configuration is run on the test split once. The FlashDot loss experiments and the preliminary re-ranking result are single runs (seed 0) on the development split; they are diagnostic. Main FlashDot results use three seeds (0, 1, 2) on the test split and report the mean and standard deviation over seeds; VAS results use one seed, because each fine-tuning arm takes several hours on our hardware. Comparisons between two systems on the same clips use a paired bootstrap over clips (10,000 resamples) of the per-clip difference, computed separately for each seed and reported as a 95% percentile interval and a two-sided p value, checked against a paired sign-flip permutation test. Seeds are the unit of replication: we report each seed's comparison rather than pooling clips across seeds. The window width $\sigma = 1$ was fixed in advance as the default for test runs; the window comparison on the test split is exploratory and not corrected for multiple comparisons.

5 Results

5.1 Which metrics respond to known delays

A timing metric should worsen steadily as a known delay grows and should rank a clip with the right kind of sound but unrelated timing as badly aligned. We delay each of 200 FlashDot development spectrograms by 2 to 64 mel frames (23 to 743 ms), filling the start with the clip's quietest frame, and compare it with the original; as a second reference we compare each clip with a different clip (Figure 2; values in Table 4).

On FlashDot, where each delay is a pure shift of identical audio, the envelope lag recovers the injected delay exactly. Onset F1 is flat up to 93 ms, inside its tolerance, and drops to the level of an unrelated clip (about 0.14) or below from 186 ms. Envelope correlation falls from 0.75 at 23 ms to about zero by 93 ms, so it cannot tell 100 ms from a second. DTW grows with delay, from 69 at 23 ms to 1968 at 743 ms, but an unrelated clip scores 10582: a large delay looks five times closer to the truth than a clip with different content. This is expected, since DTW tolerates time shifts by construction, and it makes DTW unsuited as a timing metric. Mel L1 jumps at the smallest delay and then barely separates a 743 ms delay (0.052) from an unrelated clip (0.054). We therefore use |lag|, within-one-frame, onset F1 and, on FlashDot, tone F1 as timing metrics, and leave DTW out. These delays are exact shifts of identical audio, so the check shows how each metric responds to a pure delay and does not establish how well it tracks timing errors in generated audio.

Table 1: FlashDot. Top: references. Middle: PAL as a loss and the CE baselines, trained from scratch (development split, 2000 steps unless stated). Bottom: PAL as a reward. Sync acc. is the held-out critic at tolerance 1 (chance ≈ 0.14); within 1 frame is the share of clips with |lag| within one video frame. Mean ± std over three training seeds; † single training seed. Best of 8 on the development split uses training seed 0.

Model	Split	Tone F1 ↑	|Lag| (ms) ↓	Within 1 frame ↑	Sync acc. ↑
VQGAN reconstruction	dev	0.970	2.7	1.000	1.000
VQGAN reconstruction	test	0.971	4.9	0.995	1.000
CE (teacher forcing)	dev	0.414 ±0.007	410 ±14	0.108 ±0.020	0.200 ±0.030
CE, corrupted prefix ($p_{\text{keep}} = 0.5$)	dev	0.501 ±0.028	317 ±15	0.190 ±0.017	0.267 ±0.033
CE + PAL, teacher-forced (3000 steps)	dev	0.315 ±0.016	485 ±26	0.050 ±0.015	0.112 ±0.033
CE + PAL, half-corrupted PAL prefix	dev	0.255 ±0.024	502 ±24	0.055 ±0.020	0.117 ±0.006
CE + PAL, fully corrupted PAL prefix†	dev	0.279	491	0.060	0.125
CE, corrupted prefix, best of 8†	dev	0.535	228	0.480	0.645
CE, corrupted prefix	test	0.498 ±0.014	340 ±12	0.198 ±0.016	0.233 ±0.016
best of 8	test	0.567 ±0.027	239 ±14	0.422 ±0.026	0.705 ±0.031
self-training, random selection	test	0.449 ±0.012	384 ±21	0.128 ±0.033	0.178 ±0.033
self-training, PAL selection	test	0.458 ±0.029	359 ±24	0.133 ±0.010	0.185 ±0.020

5.2 PAL as a training loss does not improve timing

On FlashDot, every way of adding PAL to the loss lowers timing quality (Table 1, middle; means over three training seeds on the development split, except for the fully corrupted row, which is one seed). With teacher-forced logits, tone F1 falls from 0.414 to 0.315 and held-out sync accuracy (tolerance 1) from 0.200 to 0.112, below chance (about 0.14), even with 1000 more training steps. Computing PAL from a half or fully corrupted prefix, so that the video is the only source of timing, does no better (tone F1 0.255 and 0.279). Corrupting the prefix of the cross-entropy pass alone, with no critic, helps instead: tone F1 0.501, |lag| 317 ms. The two diagnostics of Section 3.6 are consistent with the shortcut account below. They do not exclude other causes of the drop, such as the single loss weight ($\beta = 1$) or interference between the two gradients.

Exposure bias. The CE model does use the video under teacher forcing: over the 200 development clips, and as a mean over three training seeds, development cross-entropy is 1.54 with each clip's own visual features and 1.89 with another clip's. With a fully corrupted prefix it is 4.51 against 4.66, close to the uniform level of 4.85, so the video alone tells the model little about which code comes next. This suggests that teacher-forced predictions inherit their timing from the aligned prefix and look aligned to the critic whatever the model does with the video, while the model's free-running samples have low timing accuracy (held-out sync accuracy 0.200 at tolerance 1, against a chance level of about 0.14). A PAL gradient on teacher-forced logits would then reward copying the prefix, which the model already does.

The critic can be exploited. Corrupting the PAL prefix removes the shortcut through the prefix but opens another one. In the free-logit attack (Figure 3), optimizing logits directly against the mel critic reaches a PAL of 0.98, about that of the ground-truth codes (0.95; random codes 2.12). The decoded spectrograms are broadband texture with tone F1 0.28, barely above random codes (0.20) and far from the ground truth (0.98). Critics that see only 8, 4 or 1 band energies are fooled the same way. Coarser critics do align energy changes better at the optimum, with |lag| of 151 ms (mel), 147 ms (8 bands), 48 ms (4 bands) and 15 ms (1 band) against 644 ms for random codes, but none of them produces the tone: tone F1 at the optimum is 0.28, 0.28, 0.11 and 0.05. A critic trained only on real audio has not seen spectrograms like these, which is a likely reason it scores them as aligned.

5.3 PAL as a reward on samples

Used to choose among the generator's own samples, the same critic helps. Keeping the best of eight free-running samples from the corruption-trained CE model (candidates scored with $\sigma = 0$) raises tone F1 from 0.499 to 0.535, the share of clips within one video frame from 0.17 to 0.48, and held-out sync accuracy (tolerance 1) from 0.245 to

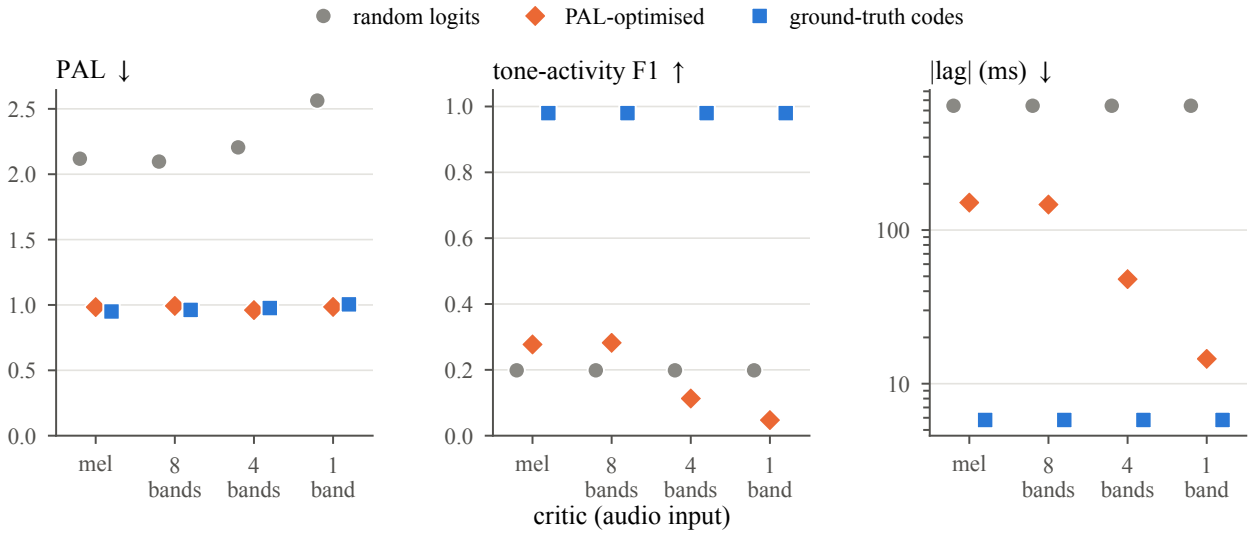


Figure 3: Free-logit attack on four FlashDot critics (eight development clips). Optimizing logits against PAL (orange) reaches the PAL of the ground-truth codes (blue) for every critic, while tone F1 stays far below the ground truth (0.98): 0.28 for the mel and 8-band critics against 0.20 for random codes (grey), and lower for the 4-band and 1-band critics. Coarser critics bring the envelope lag down, but none recovers the tone.

0.645 on the FlashDot development split (training seed 0) (Table 1). The tone F1 gain is modest; the larger held-out gain partly reflects that the two critics share features and model family. Figure 4 shows one development clip: the clip whose |lag| gain from re-ranking is the median over clips, so it shows a typical case rather than the best one.

On the FlashDot test split, over three generator seeds and with candidates scored at $\sigma = 1$ (the setting used for VAS and for self-training), best-of-8 re-ranking raises tone F1 from 0.498 to 0.567, the share of clips within one frame from 0.20 to 0.42, and held-out sync accuracy (tolerance 1) from 0.23 to 0.71 (means over seeds). Seeds are the unit of replication, so we report each seed. Against one sample, best of 8 raises tone F1 by 0.054 ($p = 0.001$), 0.084 and 0.068 (both $p < 0.001$) and the share of clips within one frame by 0.205, 0.195 and 0.270 (all $p < 0.001$); the sign is the same in every seed. The oracle that keeps the candidate with the best true tone F1 reaches only 0.39 held-out sync accuracy against 0.71 for PAL selection (Table 3), so the held-out score partly measures agreement with the selecting critic; tone F1 and the within-one-frame share are the primary timing measures.

Self-training does not keep this gain. After two rounds (400 clips per round, eight candidates, 300 steps), mean tone F1 falls to 0.458 with PAL selection and 0.449 with random selection, below the 0.498 of the starting model, and the share of clips within one frame falls from 0.20 to 0.13 with either rule. Per seed (0, 1, 2), the change in tone F1 against the starting model is −0.010 ($p = 0.47$), −0.052 and −0.059 (both $p < 0.001$) with PAL selection and −0.043 ($p = 0.002$), −0.056 and −0.050 (both $p < 0.001$) with random selection. PAL minus random selection is +0.033 ($p = 0.01$), +0.004 ($p = 0.75$) and −0.008 ($p = 0.54$) in tone F1, so the selection rule does not matter consistently. Because fine-tuning on the model's own samples lowered tone F1 with either selection rule, the FlashDot self-training arms say little about the critic's choice; the ground-truth-only control exists only on VAS.

On VAS (Table 2), the VQGAN reconstruction reaches FID 6.0 and MKL 1.03, matching the published codebook numbers (6.0 and 1.0), which checks our port of the evaluation. The pretrained SpecVQGAN transformer (top-k 100, one sample per clip) gives FID 27.8 and MKL 5.81 (published 25.4 and 5.9). Its timing is close to chance: envelope correlation 0.003, |lag| 502 ms, 5.6% of clips within one frame, and held-out sync accuracy at tolerance 0 / 1 of 0.061 / 0.167, against chance levels of about 0.05 / 0.14. The ground truth itself reaches only 0.285 / 0.536 (tolerance 0 / 1) under the held-out critic, which bounds what that metric can show.

Keeping the best of four samples per clip by PAL improves the timing measures that follow the envelope: |lag| falls by 46 ms (95% interval 16 to 74 ms, $p = 0.002$, paired bootstrap over the 768 clips), envelope correlation rises from 0.003 to 0.015 ($p < 0.001$) and held-out sync accuracy at tolerance 1 from 0.167 to 0.375, at four times the sampling cost. Onset F1 (+0.008, $p = 0.10$) does not change significantly, and the share of clips within one frame stays at 5.6 to 6.6% ($p = 0.41$), and FID and MKL stay close to the single-sample values. The held-out gain is the largest, and it is

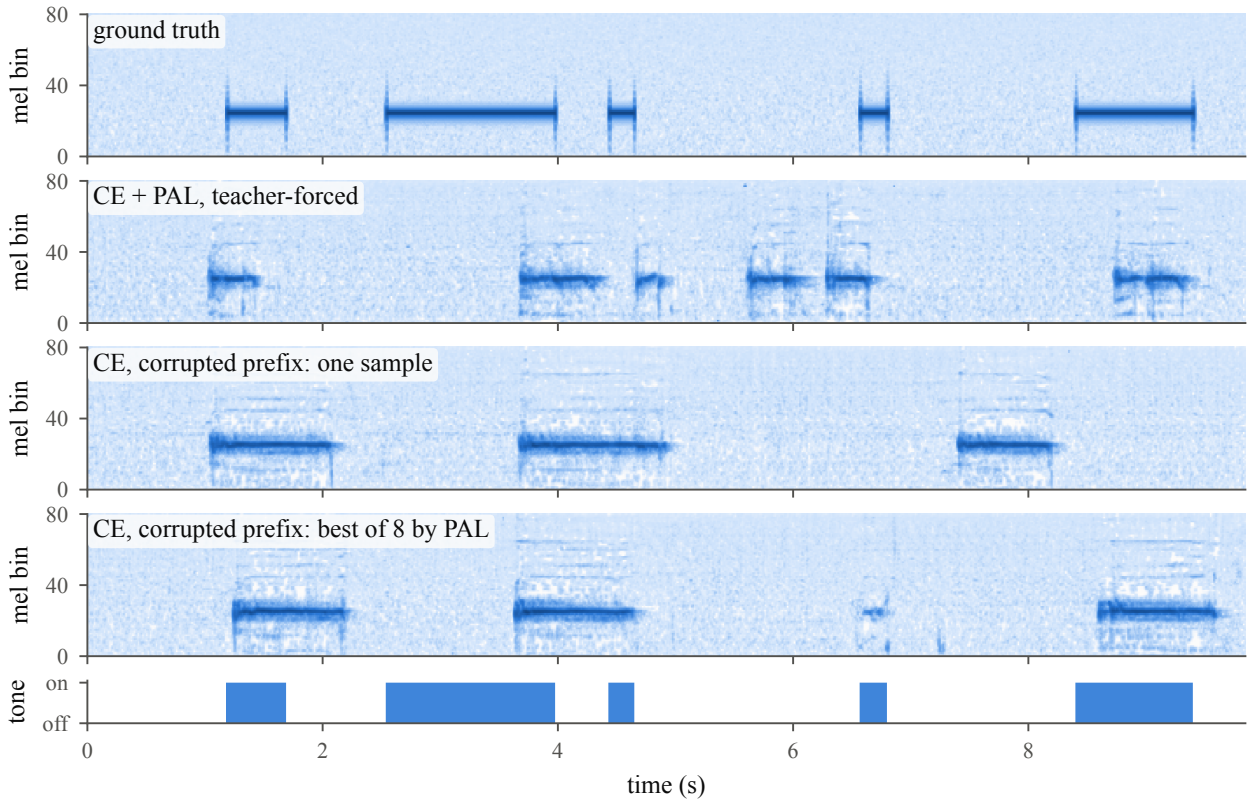


Figure 4: One FlashDot development clip (the clip with the median |lag| gain from best-of-8 re-ranking). The bottom row marks when the tone is on. The model trained with PAL as a loss (teacher-forced) adds tones that are not in the video; the corruption-trained CE model misses some tones; keeping the best of eight of its samples by PAL matches more of the tones (tone F1 0.57 on this clip, against 0.33 for the single sample), though its lag is still large (790 ms against 859 ms).

the one most open to shared bias between the two critics (Section 7).

Self-training does not carry this gain into the generator. With the same LoRA budget (two rounds of 256 clips, four candidates, 150 steps each), self-training with PAL selection, self-training with random selection and fine-tuning on ground truth alone give the same timing: no two of the three differ significantly on any timing measure (all $p > 0.1$). For PAL against random selection, onset F1 differs by +0.003 ($p = 0.50$), |lag| by +22 ms ($p = 0.13$) and sync accuracy at tolerance 1 by +0.001 ($p = 0.98$). All three lower onset F1 by about 0.02 against the pretrained model ($p < 0.001$), and only random selection lowers |lag| significantly (−35 ms, $p = 0.01$). All three also lower FID, and ground-truth-only fine-tuning lowers it most (20.4, against 24.1 with PAL selection and 25.1 with random selection). On VAS this suggests that the FID gain comes from fine-tuning on in-domain audio (FID values are single runs), and at this budget PAL selection adds nothing measurable to timing or quality.

Binding window. Table 3 scores the same eight candidates per clip under different windows. Every window improves on one sample. Tone F1 barely depends on the window (0.551 to 0.572 on FlashDot), while the other measures trade off: the within-one-frame share ranges from 0.370 to 0.463 and held-out sync accuracy from 0.590 to 0.705. On FlashDot the narrowest window ($\sigma = 0$, exact zero offset) puts more clips within one frame than $\sigma = 1$ in all three seeds (0.463 against 0.422 on average), but the difference is significant only in seed 0 ($p = 0.01$; $p = 0.36$ and 0.40 in the others). In turn, $\sigma = 1$ gives higher held-out sync accuracy at tolerance 1, the tolerance that window matches, in all three seeds (0.705 against 0.590, $p < 0.001$ in each) and higher onset F1 in all three ($p \leq 0.014$). On VAS (one run, 256 clips), $\sigma = 0$ and $\sigma = 1$ do not differ significantly on any measure. Neither asymmetric window gives a consistent gain over the symmetric $\sigma = 1$ window, and the window with a wide audio-late side ($\sigma = 0.5$, $\sigma_{\text{late}} = 2$) lowers held-out sync accuracy relative to $\sigma = 1$ on FlashDot in all three seeds ($p \leq 0.009$) and on VAS (−0.066, $p = 0.003$). The oracle

Table 2: VAS test results (768 clips of the SpecVQGAN `vas_valid` list), one sample per clip unless stated. Sync acc. is the held-out critic at tolerance 0 / 1 (chance ≈ 0.05 / 0.14). On.: onset; Env.: envelope correlation; |Lag| in ms; 1 fr.: share of clips with |lag| within one video frame. Fine-tuned rows train LoRA adapters with the same budget; each is a single training run (seed 0).

Model	On. F1 ↑	On. AP ↑	Env. ↑	|Lag| ↓	1 fr. ↑	Sync acc. ↑	FID ↓	MKL ↓
Ground truth	–	–	–	–	–	0.285 / 0.536	–	–
VQGAN reconstruction	0.813	0.778	0.918	0.2	100%	0.277 / 0.518	6.0	1.03
Pretrained SpecVQGAN	0.368	0.206	0.003	502	5.6%	0.061 / 0.167	27.8	5.81
best of 4	0.376	0.215	0.015	456	6.6%	0.151 / 0.375	26.7	5.92
GT-only fine-tune	0.346	0.188	0.006	481	5.3%	0.055 / 0.167	20.4	5.58
self-train, random	0.344	0.185	0.006	467	6.5%	0.060 / 0.165	25.1	5.95
self-train, PAL	0.347	0.186	0.004	488	6.5%	0.062 / 0.167	24.1	5.75

Table 3: Candidate selection on the test split: every row chooses among the same 8 free-running samples per clip; the "One sample" row is the first of these candidates, a separate draw from the single-sample rows of Table 1. FlashDot: corruption-trained CE generator, 200 clips, mean ± std over three generator seeds. VAS: pretrained SpecVQGAN, a 256-clip subset of the test list (to bound sampling time), one run; its FID uses these clips only and is not comparable with Table 2. σ: width of the binding window on the audio-early side, σ_{late} on the audio-late side (Equation (2)). Sync: held-out critic at tolerance 1. The oracle keeps the candidate with the best true tone F1 (FlashDot) or smallest |lag| (VAS).

	FlashDot			VAS			
Selection	Tone F1 ↑	1 fr. ↑	Sync ↑	1 fr. ↑	|Lag| ↓	Sync ↑	FID ↓
One sample	0.513 ±0.027	0.245 ±0.023	0.267 ±0.020	5.5%	525	0.164	37.0
Best of 8, $\sigma = 0$	0.551 ±0.021	0.463 ±0.034	0.590 ±0.022	7.0%	439	0.465	35.7
Best of 8, $\sigma = 0.5$	0.558 ±0.020	0.462 ±0.019	0.650 ±0.031	7.8%	432	0.465	35.9
Best of 8, $\sigma = 1$	0.567 ±0.027	0.422 ±0.026	0.705 ±0.031	7.4%	436	0.449	35.4
Best of 8, $\sigma = 2$	0.572 ±0.029	0.370 ±0.013	0.668 ±0.028	7.4%	425	0.426	35.7
Best of 8, $\sigma = 0.5$, $\sigma_{\text{late}} = 2$	0.567 ±0.024	0.380 ±0.035	0.612 ±0.015	7.8%	423	0.383	34.1
Best of 8, $\sigma = 2$, $\sigma_{\text{late}} = 0.5$	0.563 ±0.020	0.382 ±0.008	0.628 ±0.059	9.4%	422	0.426	35.2
Oracle (upper bound)	0.696 ±0.016	0.280 ±0.013	0.393 ±0.067	39.8%	97	0.227	34.2

rows show how far any PAL choice is from the best available candidate: on VAS, choosing by true lag puts 40% of clips within one frame, against 7 to 9% for every PAL window.

6 Discussion

The same frozen critic hurts timing as a loss and helps as a selector at sampling time, and the two diagnostics are consistent with one account of the difference. A differentiable loss scores whatever the gradient path produces: under teacher forcing that is a prediction already aligned by its prefix, and with the prefix removed it is a point the generator can move toward freely, including spectrograms outside anything the critic was trained on. A reward on free-running samples scores only what the generator actually produces, so the critic works inside the region where its ranking means something. That ranking did not carry over into the generator's weights: fine-tuning on the critic's choices matched fine-tuning on random choices on both datasets, and on FlashDot both fell below the starting model. The gain from selection is specific to each clip and its video, and two short rounds of cross-entropy on a few hundred chosen samples did not teach the generator to make such choices itself; larger budgets or a training signal that compares candidates directly may behave differently. This matches what is known about optimizing learned rewards (Gao et al., 2023): the less freely the optimizer can move against the reward model, the more the reward keeps its meaning. For similar setups, we suggest running the wrong-video probe and the free-logit attack before training against a critic, and using the critic to select samples at generation time. Our results give no support to the perceptual motivation for the window: no window is best on every measure, and the asymmetric windows that follow the human tolerance for late audio bring no

gain. We keep the window as a parameter of PAL rather than as a claim about perception.

7 Limitations

- **Base model.** SpecVQGAN is far from current V2A systems such as MMAudio (Cheng et al., 2025). Our results concern a codebook transformer with a frozen decoder; we make no claim about flow or diffusion models and no state-of-the-art comparison.
- **Scope of the negative result.** We observe that PAL fails as a loss for this generator with straight-through decoding, teacher-forced or corrupted prefixes, and our critics. PAL as a loss was tested only on FlashDot, with a 6-layer transformer trained from scratch and a single weight $\beta = 1$. Other gradient estimators, critics trained on generated audio, or losses on free-running samples could behave differently.
- **Compute.** All training ran on one laptop GPU, so on VAS we fine-tune LoRA adapters rather than the full transformer, and self-training uses two rounds with few candidates per clip. Larger budgets could change the size of the effect in either direction.
- **Feature-based critics and evaluation.** The critics see precomputed visual features at about 21.5 frames per second, so one offset step is about 46 ms and finer errors are invisible to them. On VAS their development offset accuracy is only about 0.12. The held-out critic uses the same features and model family as the training critic, so it is not fully independent of it. We did not compute a score from a stronger synchronization model such as Synchformer (Iashin et al., 2024), which needs raw video frames rather than the precomputed features we have; such a score would be a more independent check.
- **Binding window.** The Gaussian window, symmetric or wider on the audio-late side, simplifies a human window whose width and shape vary with stimulus, task and observer (Stevenson et al., 2012; Takeshima, 2021); it is not fitted to human data.

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A FlashDot example and metric check values

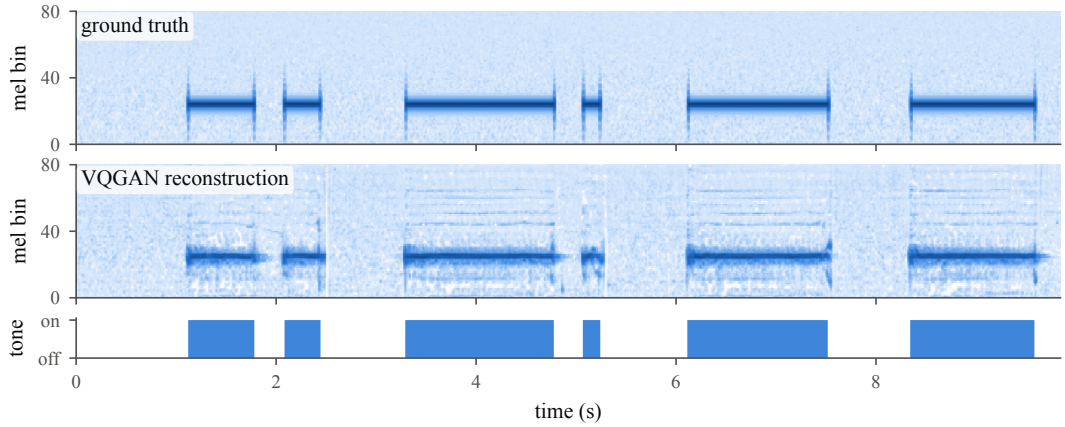


Figure 5: A FlashDot clip: ground-truth spectrogram, its VQGAN reconstruction with the public VAS codebook, and the tone label. The reconstruction keeps the timing (tone F1 0.970 over the development split) and adds codebook texture, so it bounds what any code model with this decoder can reach.

Table 4: Response of candidate metrics to known delays, mean over 200 FlashDot development clips. Arrows give the direction of a better score.

Comparison	DTW ↓	Mel L1 ↓	Onset F1 ↑	Env. corr. ↑	|Lag| (ms) ↓
Delay 0 ms	0	0.000	1.00	1.00	0
Delay 23 ms	69	0.032	1.00	0.75	23
Delay 46 ms	138	0.035	1.00	0.32	46
Delay 93 ms	271	0.036	1.00	0.01	93
Delay 186 ms	527	0.040	0.05	0.02	186
Delay 372 ms	1034	0.045	0.13	−0.01	372
Delay 743 ms	1968	0.052	0.14	−0.04	743
Other clip	10582	0.054	0.14	0.01	470

B Hyperparameters

Table 5: Hyperparameters. Values are those of the reported runs.

Component	Setting	Value
Training critic	encoder blocks / kernel / channels	2 / 5 / 256
	embedding dimension, normalization	128, L2 per frame
	group norm groups, mel pooling	8, $\times 4$
Held-out critic	blocks / kernel / channels, seed	3 / 3 / 384, 1
	audio input (FlashDot / VAS)	mel / 8 bands
Stage 1	K, τ, σ	10, 0.1, 0
	S (FlashDot / VAS)	4 / 8
	reduction	$-\frac{1}{S}\sum_s \log W_s$
	probability of ground-truth audio	0.3
	optimizer	AdamW, lr 10^{-3}, wd 10^{-4}
	steps (FlashDot / VAS), batch	2000 / 3000, 32
	selection	lowest dev loss
PAL as a loss (FlashDot)	K, S, τ, σ	8, 4, 0.1, 1
	reduction	$-\log \frac{1}{S}\sum_s W_s$
	decoding	straight-through
	window columns c, margin m	12, 1
	examples with PAL per batch	4 of 8
	PAL weight β	1
	p_{keep} (CE pass), $p^{\text{PAL}}_{\text{keep}}$	$\{1, 0.5\}$, $\{1, 0.5, 0\}$
	optimizer	AdamW, lr 3×10^{-4}, $\beta = (0.9, 0.95)$, wd 0.01
	schedule, gradient clipping	200 warm-up steps, cosine decay; 1.0
	steps, batch	2000 (3000 teacher-forced PAL), 8
FlashDot GPT	layers / width / heads	6 / 384 / 6
Self-training	rounds, steps per round	2, 300 (FlashDot) / 150 (VAS)
	clips per round, candidates N	400, 8 (FlashDot) / 256, 4 (VAS)
	batch, ground-truth share	8 (FlashDot) / 4 (VAS), 0.5
	trained parameters	all (FlashDot) / LoRA $r = 16$, $\alpha = 16$ (VAS)
	initialization	CE, corrupted prefix (FlashDot) / public model (VAS)
	optimizer	AdamW, lr 10^{-4}, $\beta = (0.9, 0.95)$, wd 0.01
	candidate scoring	training critic, $\sigma = 1$
Sampling	top-k, temperature	100, 1.0
	re-ranking samples N (FlashDot / VAS)	8 / 4
Seeds	FlashDot / VAS	3 (test) / 1

C VAS sample

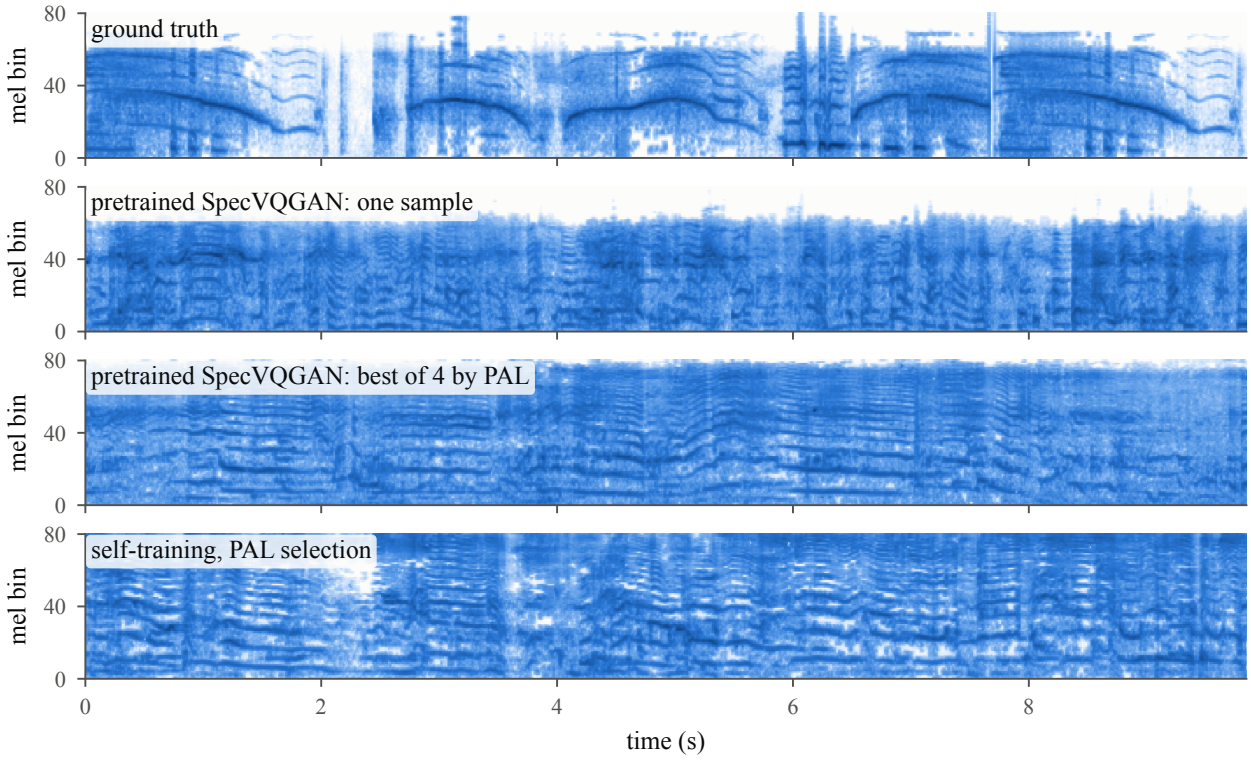


Figure 6: One VAS test clip (the clip with the median |lag| gain from best-of-4 re-ranking). Its |lag| is 116 ms for the single sample and 81 ms for the best of four. VAS audio is dense, and timing differences of this size are hard to see in a spectrogram, which is why we report onset and envelope measures.